

Flyfire.

FLYFIRE TECHNOLOGY CO.,LTD

UAV PARACHUTE!



PRODUCT CATALOG

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Flyfire technology focuses on the research and development of aircraft safety system. With the goal of "creating a safe flight era", the company is committed to providing partners with the most reliable aircraft protection equipment and eliminating risks for UAV applications.

The passive safety scheme of Flyfire technology greatly reduces the injury and loss of aircraft after the accident by means of parachute, air-assist, emergency power-off, and more Together with global partners, Flyfire leverage its expertise and innovation, provides partners with the most reliable UAV safe flight scheme, and jointly create an era of safe flight.

LAYOUT OF 51 PATENTS



2015

Relying on the military to civilian technology of the University, a university student team based on UAV safety equipment research and testing was established.

2018

Harbin Flyfire technology was established, the first generation of UAV parachute Manti came out, and reached an agreement with Harbin railway public security to protect the safety of UAVs patrolling the line.

2019

After stopping in Guangzhou, Guangzhou Flyfire technology was established and released the second generation of consumer UAV parachute Manti, which received a huge response in the market.

► 2020-2022

With the emphasis on the safety of drones, we have empowered the accumulated technology to the industry and established extensive the cooperation.

Make full use of design advantages, Integrating up and downstream supply chain capabilities and produce products with leading performance.

PRODUCT: (PARACHUTE)

OWL M350



dji CUSTOMIZED PRODUCTS BASED ON DJI PSDK

PRODUCT SPECIFICATION (OWL INDUSTRIAL PARACHUTE PRODUCT BASIC SPECIFICATIONS)

Sensor		Software		Connection to aircraft		Parachute Cloth Type		Degree of protection	
Double: IMU /barometer		APS 3.0		PSDK		Square		IPX5	
Reaction time	Working Temperature	Parachute opening time	Parachute cloth area	Affect endurance time	Weight	Alarm	Propeller stop protection	Deceleration	Max load
0.5s	-20°C~50°C	0.3s	7.29m ²	5mins	800g	Double horn	Yes	3.5m/s	9kg

ADAPTABLE MODEL : DJI Matrice 300 series/350 series



- 1:Automatic and manual dual-mode parachute opening.
- 2:IPX5 protection grade, maintenance free for two years under normal environment.
- 3:Parachute opening in 1.3 seconds,3.5m/s safe descent speed.

PARACHUTE PARAMETERS:

Model : OWL M30 Pro
Adaptable Model : DJI-Matrice 30 series
Maximum Load : $\leq 4.5\text{kg}$
Sink Rate : 4.6m/s
Size Of The Parachute Cabin: $130 \times 105 \times 45\text{ mm}$
Area Of Parachute Fabric : 2.25m^2
System Weight : 280g
Type Of Parachute : SQUARE Parachute
Working Temperature : $-20^{\circ}\text{C} \sim 60^{\circ}\text{C}$
Working Altitude : $\leq 3500\text{m}$
IP Grade : IP45
Wind Resistance Rating : ≤ 7
Response Time : $500\text{ms} \sim 600\text{ms}$
Response Distance : $8 \sim 15\text{m}$

PRODUCT: (PARACHUTE)

OWL-M30^{PRO}



dji CUSTOMIZED PRODUCTS BASED ON DJI PSDK



Sink Rate : 4.6m/s



Firmware Upgrade



Mutual Interaction



Audible And Visual Alarm



Reuse



Quick Open



Sensor Redundancy



Quick Loading System

PRODUCT: (PARACHUTE)

manti III

PARACHUTE PARAMETERS:

Compatible models: Mavic Pro\2\3\4\Mavic Air 2\3\Phantom 3\4

Upgrade point : Replaceable cabin \ Firmware DFU upgrade \ 3 - 3plus Parachute bag \ Buzzer \ Support Mavic - Mavic Air - Phantom series.

Protection height : Protection height $\geq 25\text{m}$, The rate of major aircraft explosion accidents decreased to 0.03%.

Working Temperature : $-20^{\circ}\text{C} \sim 60^{\circ}\text{C}$

Reaction time : 500ms

Normal working hours : 4hours

INDUSTRIAL CHIP, 0 ERROR

- 1 The crash began to fall
- 2 Stop slurry
- 3 Manti automatic parachute opening
- 4 Rescue aircraft



IMU / Barometer / APS System

Accurately Identify 8 Major Accidents

- 1: Signal interference.
- 2: Branch failure.
- 3: Wire obstruction.
- 4: Picture transmission black screen.
- 5: Low charge.
- 6: Signal loss.
- 7: Aircraft out of control.
- 8: Wind interference.

Firmware DFU upgrade

Upgrade the algorithm to be compatible with new models



PRODUCT: (PARACHUTE)

mantiV



1. REUSABLE
2. FIRMWARE UPDATE

Sink Rate : 5.5m/s



Manti 4Dock Emergency Light Edition | Manti 4Dock Basic Edition(No Lights) | Manti4



Models: Manti 4

Compatible models: MATRICE4\4T\Mavic Pro\2\3\4\Mavic Air 2\3\Phantom 3\4

Communication: -

Sink rate: 5.5m/s

Working hours: 4.5h

Response distance: 8m

Working altitude: ≤4000m

Impact kinetic energy: 18.15Joules

Product dimensions: 60*60*45mm

Weight: 72g

Canopy area: 1.1m²



PRODUCT: (PARACHUTE)
mantiV

Sink Rate: 5.8m/s

dji CUSTOMIZED PRODUCTS BASED ON DJI PSDK



Manti 4Dock Basic Edition(No Lights)



Manti 4Dock Emergency Light Edition
Emergency Light Control:



DJI Pilot 2 | Flighthub 2

Models: Manti 4Dock Emergency Light Edition/Manti 4Dock Basic Edition(No Lights)

Compatible models: Matrice 4D/4TD

Communication: eport-lite PSDK

Function: 3+3W Red and Blue Strobe/ FTS

Software control: DJI Pilot 2 / flighthub-2

Sink rate: 5.8m/s

Working hours: 4.5h

Response distance: 8m

Working altitude: $\leq 4000\text{m}$

Impact kinetic energy: 29.241Joules

Product dimensions: 68*60*54mm(E)/68*60*47mm(B)

Parachute compartment dimensions: 60*60*32mm

Weight: 92g(Emergency Light Edition)/80g(Basic Edition(No Lights))

Canopy area: 1.1m²



PRODUCT: (AIRBAG)

DRONE BUOY

**SAFETY DESIGN FOR DRONES
OPERATING OVER WATER**



Product Name: Flyfire Drone Water Floating Device

Compatible Models: Universal

Activation Method: Electronic Trigger

Product Weight: 140g

Supported Floating Aircraft Weight: 3.5 - 6kg

Activation Time: 1S trigger, 3S floating

Applicable Temperature: -20°C to 60°C

Product Dimensions: 154x40x43mm

Floating Time: ≥24h

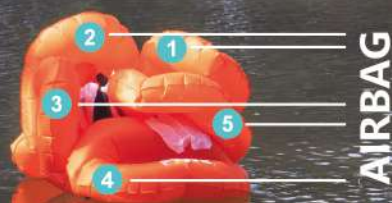
Gas Displacement: 6000cc

PRODUCT: (AIRBAG)

DRONE BUOY



FPD 55 *Drone Water Floating Device*



Product Name: Flyfire Drone Water Floating Device (55kg)

Model: FPD55

Activation Time: 1 second

Shelf Life: 1 year

Activation Method: Water sensor trigger

Compatible Models: Universal

Maximum Takeoff Weight for Compatible Drones: 55kg

PRODUCT: (AIRBAG)
FLYFIRE ACTIVE RESCUE BUOY



**ENDOW DRONES WITH THE CAPABILITY FOR
WATER SURFACE RESCUE**

ADAPTABLE MODEL

Product Name: Flyfire Active Rescue Buoy
Net Weight: $\leq 300\text{g}$
Gas Displacement: 14500cc
Buoyancy: 14.5kg
Battery Type: Button cell
Battery Life: 1 year
Cylinder Shelf Life: 1 year
Activation Method: Electronic trigger
Activation Time: 3s trigger, 6s floating
Static Dimensions: $52\text{mm} \times 178\text{mm}$
Deployed Dimensions: $684\text{mm} \times 480\text{mm} \times 120\text{mm}$
Mounting Method: Suspended by dropper
Gas Retention: $\leq 5\%$ gas loss within 24 h





PRODUCT: (PARACHUTE) UPF CONTROLLER

Product Name: UPF Safety Parachute Controller
Product Model: UPF Controller
Communication Method: UAVCAN
Parachute Deployment Method: Protocol deployment
 APS automatic deployment
Power Supply: 12v
Response Time: 500~700ms
Black Box Function: Yes
Protection Level: IP45
Operating Temperature: 0°C~55°C
Operating Altitude: ≤4000m
Product Dimensions: 71*55*19mm
Weight: 54g
Manufacturer: Flyfire Technology Co., Ltd.



FLYFIRE PARACHUTE OPEN JUDGMENT SYSTEM



UPF PRODUCT: (PARACHUTE) DRONE PARACHUTE



Universal parachute platform for unmanned aerial vehicles weighing **15kg ~ 150kg**.



Communication Protocol:	PWM\UAVCAN
Sink Rate:	4.5~5.5m/s
Remote parachute opening:	supported
Reusable:	supported
Technical consultant:	supported

Please contact us for more detailed information: info@xflyfire.com

UPF **PRODUCT: (PARACHUTE)** DRONE PARACHUTE

ADAPTABLE MODEL

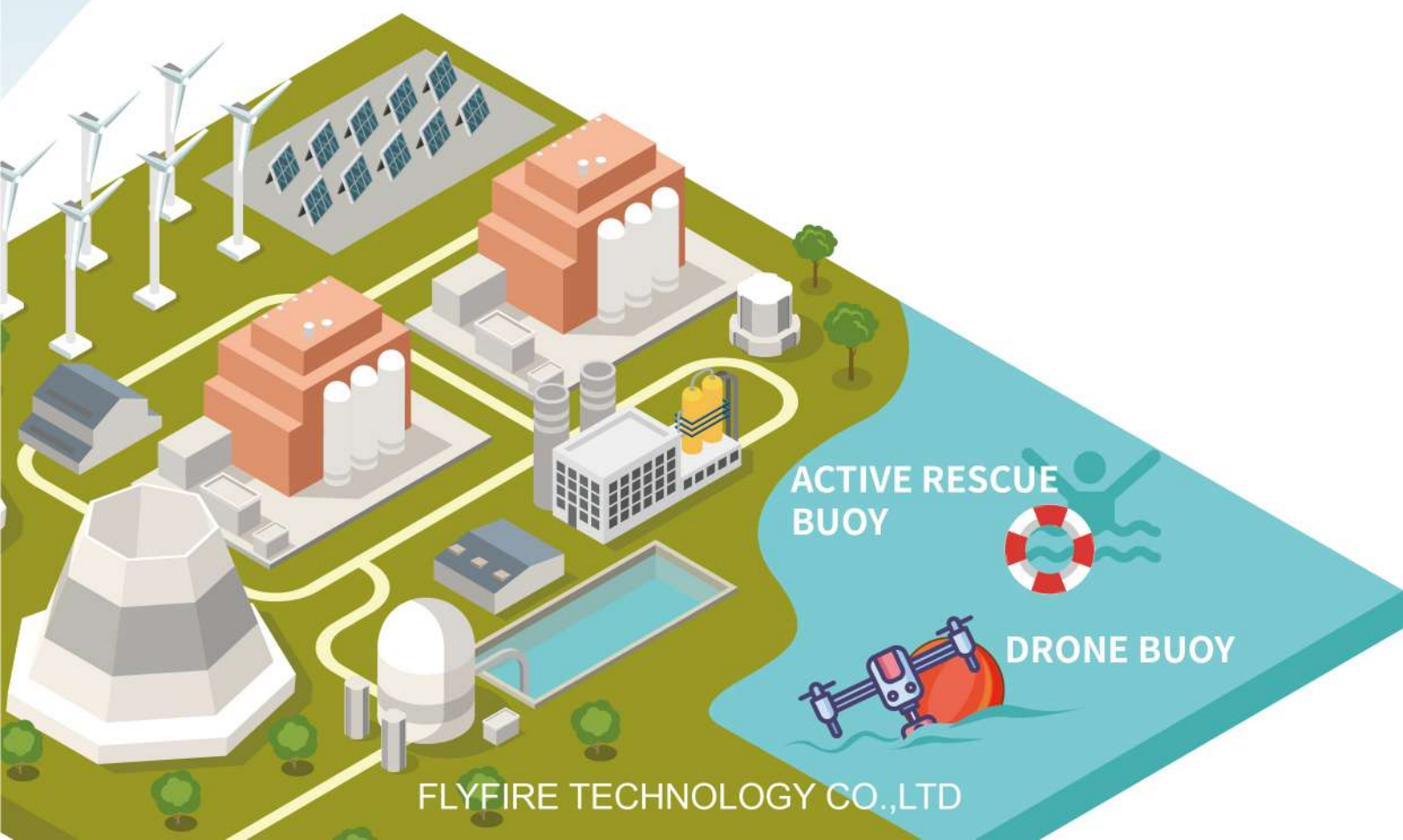
Product Model	Compatible Aircraft Type	Rated Payload	Product Weight	Canopy Area	Canopy Bay Dimensions (mm)	Stable Descent Speed	Response Time	Response Distance	Canopy deployment time	Minimum Deployment Altitude
UPF15	Multirotor UAV	15 kg	≤700 g	7.2 m ²	180×180×90.5	4.5 m/s	0.5 s	1.25 m	3.0 s	30 m
UPF30	Multirotor UAV	30 kg	≤820 g	10 m ²	180×180×130	5.5 m/s	0.5 s	1.25 m	3.5 s	40 m
UPF35	Multirotor UAV	35 kg	≤1120 g	13.5 m ²	200×200×123	5.5 m/s	0.5 s	1.25 m	4.0 s	40 m
UPF75	Multirotor UAV	75 kg	≤1600 g	17 m ²	318×190×100	4.8 m/s	0.5 s	1.25 m	3.4 s	40 m
UPF100	Multirotor UAV	100 kg	≤2050 g	25 m ²	318×190×100	5.5 m/s	0.5 s	1.25 m	3.5 s	≥60 m
UPF150	Multirotor UAV	150 kg	≤3000 g	37 m ²	210×210×265	≤5.5 m/s	0.5 s	1.25 m	3.5 s	98 m



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ACTIVE RESCUE
BUOY



DRONE BUOY



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